

IMPACT CRATERS

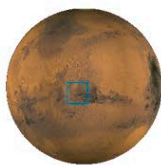
The Martian surface is scarred by tens of thousands of craters, about 1,000 of which have been given names. They range from simple bowl craters less than 3 miles (5 km) across to basins hundreds of miles wide. The oldest craters are found in the southern hemisphere and have been eroded throughout their lifetimes. Their floors have been filled and their rims degraded, and the craters have become characteristically shallow. Smaller, fresher-looking craters have formed on top of them. The ejecta has been distributed by flowing across the surface rather than being flung through the air.

ANCIENT GEOLOGICAL FEATURE

Large impact craters such as Hale (right) have had their central peaks and terraced walls continuously eroded for up to 4 billion years.

MERIDIANI PLANUM

Victoria Crater



TYPE Crater

AGE Under 100 million years

DIAMETER 0.5 miles (800 m)

Victoria is a small impact crater about two-thirds the size of the Arizona meteorite crater on Earth (see p.221). Victoria's beautifully scalloped edges have been eroded by winds, gradually increasing its diameter, and like many Martian craters, its floor is covered with dunes of wind-blown dust. The



CAPE ST. VINCENT

At Cape St. Vincent, a rocky outcrop on the northern rim of Victoria, layers of bedrock are topped by looser material thrown out by the impact that formed Victoria.

crater was explored by the Mars rover Opportunity over 1 Martian year (or 2 Earth years, from 2006 to 2008).

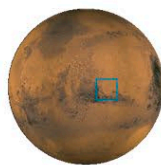
Half of that time was spent driving along part of the crater's rim before it carefully edged down a slope into the interior at an opening called Duck Bay. For the next Earth year, it examined rocky outcrops along the crater's walls with the instruments on its robot arm, finally driving out again to resume its trek across the Martian surface.

DUNE-FILLED CRATER

Rippling sand dunes cover the floor of Victoria, as seen in this enhanced-color view from the Mars Reconnaissance Orbiter.

ARABIA TERRA

Schiaparelli Crater



TYPE Large crater

AGE About 4 billion years

DIAMETER 293 miles (471 km)

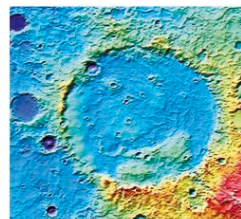
This crater takes its name from the astronomer Giovanni Schiaparelli (see p.220), who spent much of his working life studying Mars. It is a highly circular crater, as are most Martian craters, although a significant number are elliptical—a rarity on the Moon and Mercury. Schiaparelli straddles the equator and is the largest crater in the Arabia Terra. It is an old crater formed by an impacting body when the planet was young and shows signs of degradation. The rim has been smoothed down and in parts is completely missing. Any central peak in the crater has been obliterated. Material has been deposited within the crater, and smaller craters have formed across the whole area. Wind continues to shape the landscape by erosion and by moving surface material.

WIND EROSION

These layers of ancient rock sediments on the floor of an impact crater lying within the northwestern rim of Schiaparelli have been eroded and exposed by the wind.

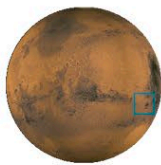
SHALLOW CRATER

Here, color is used to indicate altitude. The crater floor is at the same height as much of the surrounding terrain. Higher deposits are in green. The degraded rim (yellow) is only about 0.75 miles (1.2 km) above the floor.



TERRA TYRRHENA

Huygens Crater



TYPE Multiringed crater

AGE About 4 billion years

DIAMETER 292 miles (470 km)

Huygens is one of the largest impact craters in the heavily cratered southern highlands of Mars. It was formed during the period of intense bombardment within the first 500 million years of the planet's early history. The age of craters such as Huygens is determined by counting the number of craters that overlay their rims. Huygens has a second ring inside its mountainous rim. This has been filled by material carried into the ring. The rim is heavily eroded, and markings on it suggest that surface water has run off it at some



MESAS

Smooth-topped hills (mesas) on the crater floor are left behind as a former smooth layer of material is eroded to reveal a more rugged surface.

time. The pattern of markings is reminiscent of dendritic drainage systems on Earth, which from above

look like the trunk and branches of a tree. Dark material within this crater's drainage channels was carried either by the draining water or by the wind.

EASTERN RIM

In this perspective view across Huygens's eastern rim (foreground) to the surrounding terrain, a branchlike network of drainage channels flows away from the rim, and small, more recently formed craters can be seen.

